Department of Physics, IIT-Kanpur

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Homework-7

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1. Using the Frobenius method, obtain two linearly independent solutions of the following differential equations:

   (i)
   $$\left[x^2\frac{d^2}{dx^2}+x\frac{d}{dx}-n^2\right]y(x)=0,$$
   where n is any number. This equation is obtained from two-dimensional Laplace's equation.

   (ii)
   $$\left[x^2\frac{d^2}{dx^2}+C_1x\frac{d}{dx}+x^2+C_2\right]y(x)=0,$$
   for two different cases: (a) $C_1=4, C_2=2$ and (b) $C_1=2, C_2=-2$.

   (iii)
   $$\left[\frac{d^2}{dx^2}+x\frac{d}{dx}+1\right]y(x)=0.$$

   (iv)
   $$\left[\frac{d^2}{dx^2}-\frac{n(n+1)}{x^2}\right]y(x)=0,$$
   where n is non-negative real number.

2. In the class, we have derived $J_1(x)$ and $J_{1/2}(x)=\sqrt{2/(\pi x)}\sin x$. Using Wronskian method, obtain the second independent solution $N_1(x)$ (at least first three terms in the series) and $N_{1/2}(x)$.

3. A quantum particle of mass M is confined in a cylinder of radius R and height H. The potential is described as
   $$V(\rho,\phi,z) = 0, \quad 0<\rho<R, 0<z<H \tag{1}$$
   $$= \infty \quad \text{otherwise.} \tag{2}$$
   Obtain the discrete energy spectrum and the corresponding wave functions.

4. In class we have seen that two-diemensional plane wave can be expressed as a series of cylindrical Bessel functions:
   $$e^{\pm ix\cos\phi} = \sum_{n=-\infty}^{\infty} i^n J_n(x)e^{\pm in\phi}. \tag{3}$$
   From one of the above relations, show that
   $$\cos(x\cos\phi) = J_0(x)+2\sum_{n=1}^{\infty}(-1)^n J_{2n}(x)\cos(2n\phi) \tag{4}$$
   $$\sin(x\cos\phi) = 2\sum_{n=1}^{\infty}(-1)^{n+1}J_{2n-1}(x)\cos[(2n-1)\phi]. \tag{5}$$
   From these results, express $\sin x$ and $\cos x$ as a series of $J_n(x)$.

5. **Arfken & Weber: 11.1.13**: The scattering amplitude of a quantum scattering problem is given as

$$f(\theta) = \frac{k}{2\pi} \int_0^{2\pi} \int_0^R e^{ik\rho \sin\theta \sin\phi} \rho d\rho d\phi, \tag{6}$$

where θ is the angle through which the particle is scattered and R is the radius of the scatterer. Show that

$$|f(\theta)|^2 \sim \left[\frac{J_1(kR\sin\theta)}{\sin\theta}\right]^2. \tag{7}$$

6. The Helmholtz equation in spherical polar coordinates:

$$\left[\nabla_{\mathbf{r}}^2 + k^2\right] \psi(r, \theta, \phi) = 0, \tag{8}$$

where k is a constant. For a free particle of mass M and energy E, $k^2 = 2ME/\hbar^2$. Assume $\psi(r, \theta, \phi) = R(r)P(\theta)\Phi(\phi)$.

(a) Using the separation of variables, show that

$$\left[\frac{d^2}{d\phi^2} + m^2\right] \Phi(\phi) = 0, \tag{9}$$

$$\left[\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d}{d\theta}\right) - \frac{m^2}{\sin^2\theta} + Q\right] P(\theta) = 0 \tag{10}$$

and

$$\left[\frac{d}{dr}\left(r^2\frac{d}{dr}\right) + k^2r^2 - Q\right] R(r) = 0. \tag{11}$$

Here, m^2 and Q are two separation constants.

(b) We have seen in the lectures that m must be an integer to have single-valued wave function and Q must have form like $Q = n(n+1)$ with n is non-negative integer to get converging series solution. In Eq. (9), why do we choose m^2 as a separation constant, instead of m?

Note: In physical systems, m is identified as azimuthal quantum number and n is identified as orbital angular momentum quantum number.

(c) Equation (11) is not the Bessel's differential equation. Obtain the following Bessel's differential equation by introducing a new dimensionless variable $x = kr$ and substitute $R(x) = Z(x)/\sqrt{x}$ and $Q = n(n+1)$ into Eq. (11):

$$\left[x^2\frac{d^2}{dx^2} + x\frac{d}{dx} + x^2 - (n+1/2)^2\right] Z(x) = 0. \tag{12}$$

$Z(x)$ is the Bessel function of order $n+1/2$: $Z(x) = J_{n+1/2}(x)$. Therefore, the original solution is

$$R(x) = \frac{Z(x)}{\sqrt{x}} = \frac{J_{n+1/2}(x)}{\sqrt{x}}. \tag{13}$$

The spherical Bessel function of first kind of order n is given as

$$j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+1/2}(x). \tag{14}$$

It is convention to introduce $\sqrt{\frac{\pi}{2}}$ in $j_n(x)$. Similarly, the spherical Bessel function of second kind of order n is given as

$$n_n(x) = (-1)^{n+1}\sqrt{\frac{\pi}{2x}} J_{-(n+1/2)}(x) = \sqrt{\frac{\pi}{2x}} N_{n+1/2}(x). \tag{15}$$

The zeroth-order spherical Bessel function is $j_0(x) = \frac{\sin x}{x}$. Evaluate $n_0(x)$ by using the Wronskian approach.

7. Prove the following recurrence relations for the spherical Bessel function $j_n(x)$:

(a)

$$j_{n-1}(x) + j_{n+1}(x) = \frac{2n+1}{x} j_n(x). \tag{16}$$

(Hint: use $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$.]

(b)

$$j_n'(x) = \frac{n}{2n+1} j_{n-1} - \frac{n+1}{2n+1} j_{n+1}(x). \tag{17}$$

[Hint: use $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$ and $2J_\nu'(x) = J_{\nu-1}(x) - J_{\nu+1}(x)$.]

(c)

$$j_{n\pm1}(x) = \frac{(n + \frac{1}{2} \mp \frac{1}{2})}{x} j_n(x) \mp j_n'(x). \tag{18}$$

Using $j_0(x) = \frac{\sin x}{x}$, show that

$$j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x} \tag{19}$$

$$j_2(x) = \frac{3\sin x}{x^3} - \frac{3\cos x}{x^2} - \frac{\sin x}{x}. \tag{20}$$

Plot these three functions.

(d)

$$\frac{d}{dx}\left[x^{n+1} j_n(x)\right] = x^{n+1} j_{n-1}(x). \tag{21}$$

(e)

$$\frac{d}{dx}\left[x^{-n} j_n(x)\right] = -x^{-n} j_{n+1}(x). \tag{22}$$

Note: All the recurrence relations mentioned here are valid for $n_n(x)$ also.

8. a) Substitute $x = \cos\theta$ into Eq. (10) and obtain the associated Legendre differential equation:

$$\Big[(1-x^2)\frac{d^2}{dx^2} - 2x\frac{d}{dx} - \frac{m^2}{1-x^2} + n(n+1)\Big]y(x) = 0. \tag{23}$$

Show that $x = 0$ is an ordinary point, $x = \pm 1$ and $x = \infty$ are regular singular points.

b) Consider the Legendre differential equation and assuming the ansatz for the series solution as

$$y(x) = \sum_{j=0}^{\infty} a_j x^{j+k}. \tag{24}$$

Obtain the following recurrence relation:

$$a_{j+2} = \frac{(j+k)(j+k+1) - n(n+1)}{(j+k+1)(j+k+2)} a_j. \tag{25}$$

c) Two roots of the indicial equation are $k = 0$ and $k = 1$. For $k = 0$, a_1 may or may not be zero. For $k = 1$, a_1 must be zero. For $k = 0$ case, we assume $a_1 = 0$.

For $k = 0$, show that

$$y_1(x) = a_0\Big[1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)n(n+1)(n+3)}{4!}x^4 - ...\Big] \tag{26}$$

and for $k = 1$, show that

$$y_2(x) = a_0\Big[x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - ...\Big]. \tag{27}$$

9. Construct an orthonormal set from the set of functions $u_n(x) = x^n, (n = 0, 1, 2...)$ in the interval $-1 \le x \le +1$ by using Gram-Schmidt orthogonalization method. You will see n-th such function is proportional to $P_n(x)$.

10. Express the Dirac-Delta function $\delta(x \pm 1)$ in a series of $P_n(x)$. Assume that the entire delta function is covered when integrating over $[-1, +1]$.

11. Arfken: 12.3.11: The amplitude of a scattered wave is given by

$$f(\theta) = \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)e^{i\delta_l}\sin\delta_l P_l(\cos\theta). \tag{28}$$

Here, θ is the scattering angle, l is orbital angular momentum quantum number, $\hbar k$ is momentum of the incident particle and δ_l is the phase shift produced by the scattering potential. Calculate the total cross-section

$$\sigma_{\text{tot}} = \int |f(\theta)|^2 d\Omega, \tag{29}$$

where Ω is the solid angle.

12. Show that

$$e^{i\mathbf{k}\cdot\mathbf{r}} = e^{ikr\cos\theta} = \sum_{l=0}^{\infty} i^l(2l+1)j_l(kr)P_l(\cos\theta).$$

Using the above result, show that

$$\int_V e^{i\mathbf{k}\cdot\mathbf{r}} d^3r = 4\pi R^3 \frac{j_1(kR)}{kR}.$$

Here V is the volume of a sphere of radius R.